

Customer Churn Prediction and Business Analytics

Author

Dustin Greenwood

Contents

1. Introduction	2
2. Objectives	3
3. Dataset Description.....	4
4. Methodology	5
4.1 Data Cleaning	5
4.2 Exploratory Data Analysis	5
4.3 Machine Learning Model.....	5
5. Results	6
5.1 Churn Distribution.....	6
5.2 Contract Type and Churn	7
5.3 Tenure and Churn.....	8
6. Model Performance	9
7. Feature Importance Analysis	11
8. Business Insights	12
9. Recommendations	13
10. Dashboard Development	14
11. Conclusion	15

1. Introduction

Customer churn refers to the loss of customers who discontinue a company's products or services. For subscription-based businesses such as telecommunications companies, customer retention is critical because acquiring new customers is often more expensive than retaining existing ones.

The objective of this project was to analyse customer behaviour patterns, identify factors associated with customer churn, and develop a machine learning model capable of predicting customers who are likely to leave the company. The project also aimed to provide actionable business insights through data visualisation and dashboard development.

2. Objectives

The objectives of this project were:

- To explore and understand customer behaviour patterns.
 - To identify factors contributing to customer churn.
 - To develop a predictive machine learning model.
 - To evaluate model performance using classification metrics.
 - To create a business intelligence dashboard for decision-making.
 - To provide recommendations for improving customer retention.
-

3. Dataset Description

The dataset used in this project was the IBM Telco Customer Churn Dataset.

Dataset Characteristics

- Number of observations: 7,043 customers
- Number of variables: 21
- Target variable: Churn

The dataset contains customer demographic information, service subscriptions, contract information, billing details, and customer retention status.

Key Variables

VARIABLE	DESCRIPTION
TENURE	Length of customer relationship in months
MONTHLYCHARGES	Monthly service charges
TOTALCHARGES	Total charges incurred
CONTRACT	Contract type
INTERNETSERVICE	Internet service provider type
TECHSUPPORT	Technical support subscription
ONLINESECURITY	Online security subscription
CHURN	Customer retention status

4. Methodology

4.1 Data Cleaning

The following preprocessing steps were performed:

- Missing values in the TotalCharges variable were identified and treated.
- The TotalCharges variable was converted to numeric format.
- Customer identifiers were removed from modelling.
- Categorical variables were encoded using One-Hot Encoding.
- Numerical variables were standardised through a preprocessing pipeline.

4.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to investigate customer characteristics and identify potential churn drivers.

The following visualisations were created:

- Churn Distribution
- Contract Type vs Churn
- Tenure vs Churn
- Feature Importance Analysis

4.3 Machine Learning Model

A Random Forest Classification model was selected due to its ability to handle nonlinear relationships and mixed variable types.

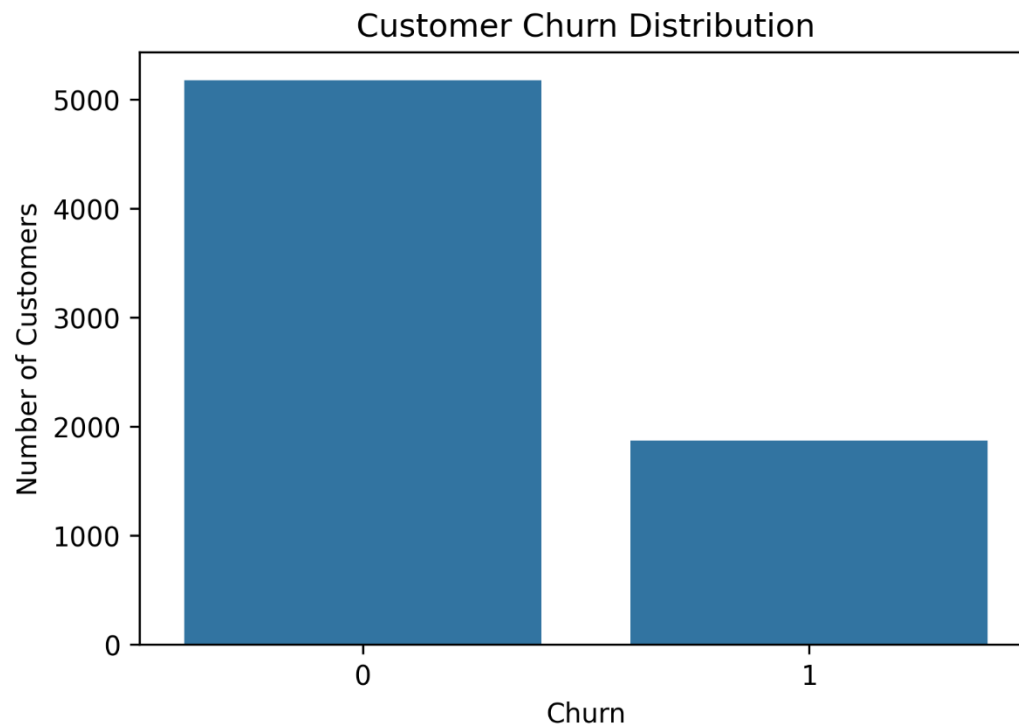
The dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

A balanced Random Forest model was used to address class imbalance.

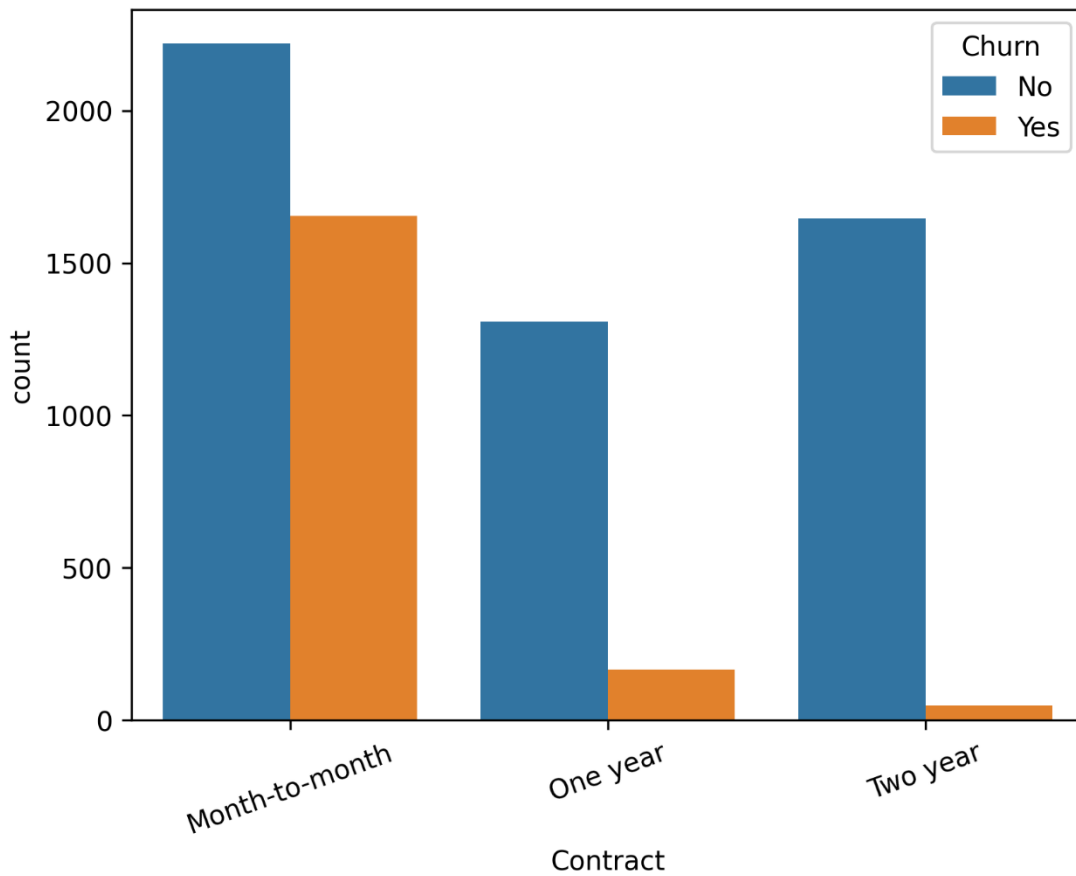
5. Results

5.1 Churn Distribution



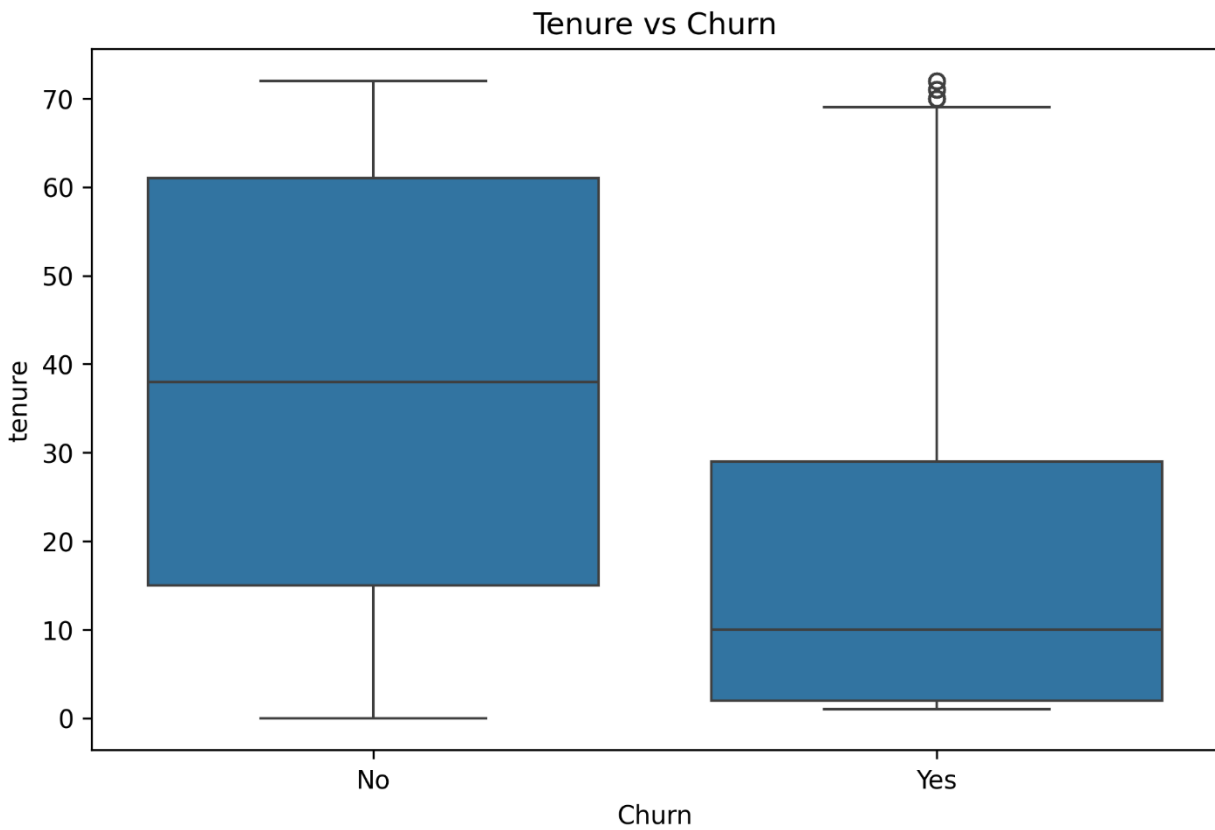
The analysis revealed that approximately 26.5% of customers had churned, while 73.5% remained with the company. This indicates a moderate class imbalance in the dataset.

5.2 Contract Type and Churn



Customers on month-to-month contracts exhibited substantially higher churn rates than customers on one-year and two-year contracts. This suggests that long-term contractual commitments improve customer retention.

5.3 Tenure and Churn



Customers who churned generally had significantly shorter tenure periods compared to retained customers. This indicates that customer attrition is most likely to occur during the early stages of the customer lifecycle.

6. Model Performance

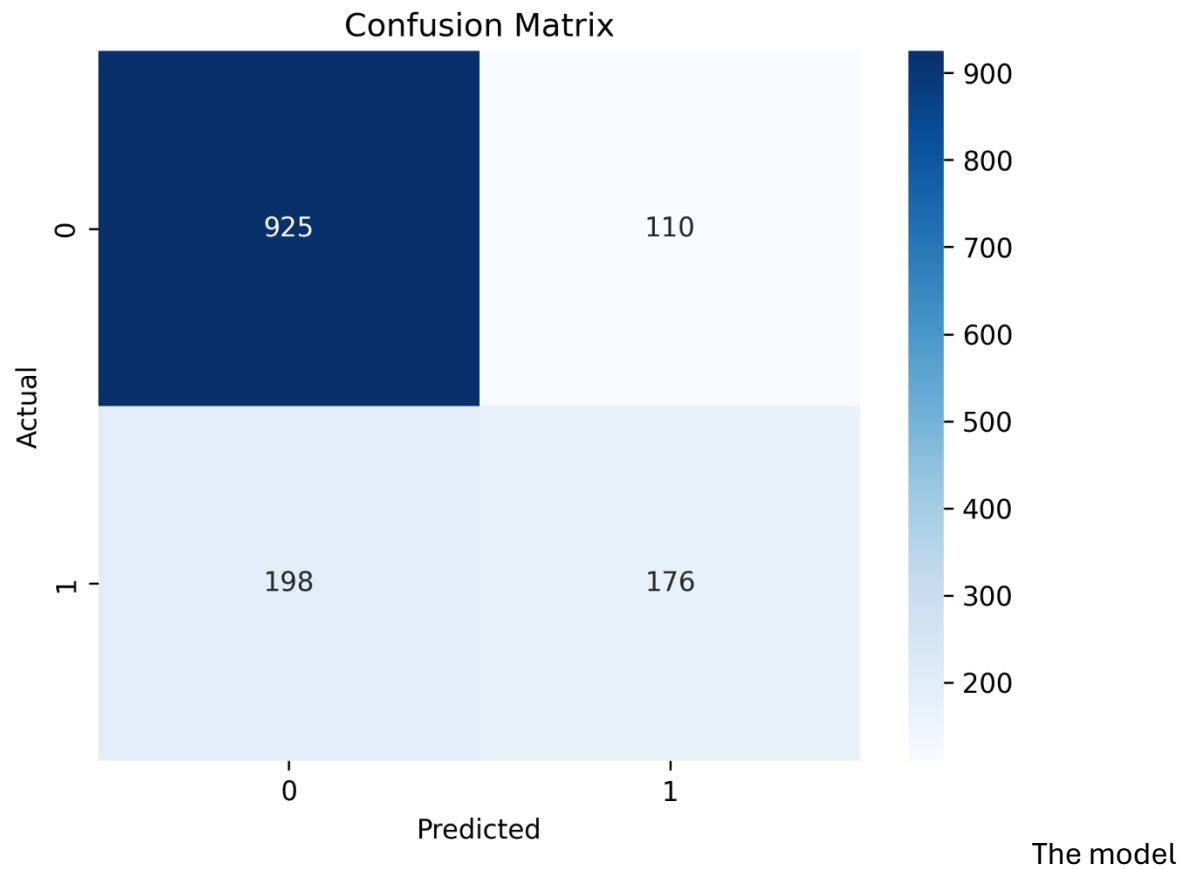
Classification Results

METRIC	VALUE
ACCURACY	78.1%
PRECISION	62%
RECALL	47%
F1 SCORE	53%

The Random Forest model achieved an overall accuracy of 78.1%, indicating good predictive capability.

Although the model performed well in identifying retained customers, churn prediction remains challenging due to class imbalance and customer behavioural complexity.

Confusion Matrix



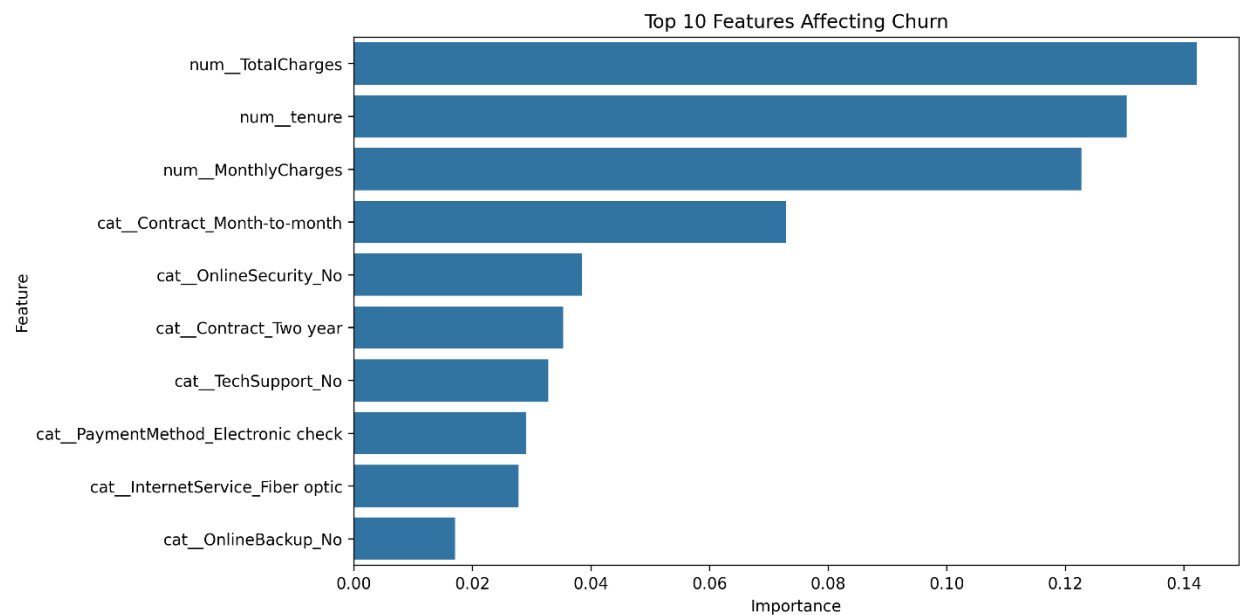
correctly classified:

- 925 retained customers
- 176 churned customers

The model incorrectly classified:

- 110 retained customers as churn risks
 - 198 churned customers as retained customers
-

7. Feature Importance Analysis



The Random Forest model identified the following variables as the most influential predictors of churn:

RANK	FEATURE
1	Total Charges
2	Tenure
3	Monthly Charges
4	Month-to-Month Contract
5	Online Security
6	Two-Year Contract
7	Technical Support
8	Electronic Check Payment

The findings indicate that customer spending behaviour, contract commitments, and service quality factors play significant roles in customer retention.

8. Business Insights

Several important business insights emerged from the analysis:

Customer Retention

Customers with shorter tenure are substantially more likely to churn.

Contract Strategy

Month-to-month customers represent the highest-risk customer segment.

Service Quality

Customers without online security and technical support services are more likely to leave the company.

Financial Behaviour

Higher monthly charges are associated with increased churn risk.

9. Recommendations

Based on the findings, the following recommendations are proposed:

Improve Customer Onboarding

Develop customer engagement programmes during the first 12 months of service.

Promote Long-Term Contracts

Offer discounts and incentives for customers to migrate from month-to-month contracts to longer-term agreements.

Expand Support Services

Encourage adoption of technical support and online security services.

Implement Predictive Monitoring

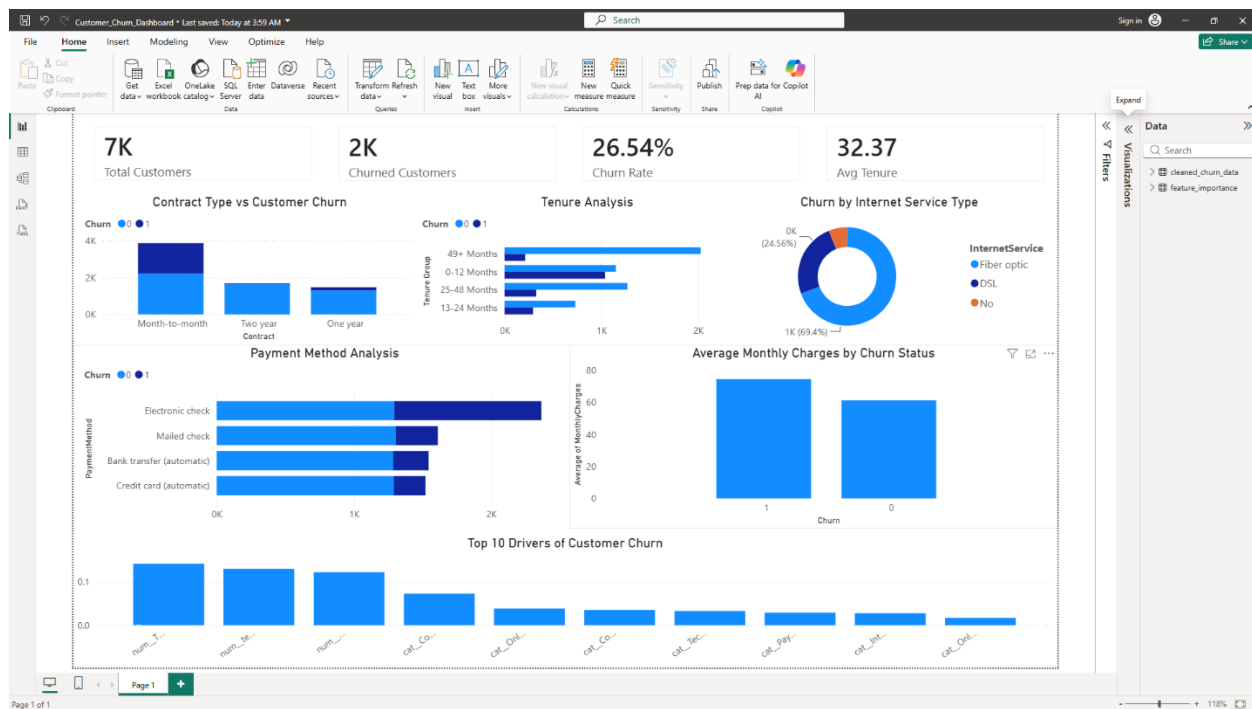
Deploy machine learning models to identify high-risk customers before churn occurs.

10. Dashboard Development

A Power BI dashboard was developed to present key performance indicators and customer churn insights.

The dashboard includes:

- Total Customers
- Churn Rate
- Average Monthly Charges
- Average Customer Tenure
- Contract Type Analysis
- Internet Service Analysis
- Payment Method Analysis
- Feature Importance Analysis



The dashboard enables stakeholders to interactively explore customer churn patterns and support data-driven decision-making.

11. Conclusion

This project successfully demonstrated the application of data analytics and machine learning techniques to customer churn prediction.

The Random Forest model achieved an accuracy of 78.1% and identified customer tenure, contract type, monthly charges, and service subscriptions as the primary drivers of churn.

The results highlight opportunities for telecommunications companies to improve customer retention through targeted interventions, improved customer support services, and strategic contract management.

Future work may include implementing advanced machine learning algorithms such as XGBoost, LightGBM, and SMOTE-based oversampling techniques to improve churn detection performance.